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Job and worker density and transit network dynamics

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ABSTRACT

This paper proposes a general framework to explore the interaction between land use and transport systems. Hypotheses about those relationships are generated. A series of statistical tests are conducted to explain the co-development of land use and transit networks for metropolitan areas at a micro-geographic scale and to disentangle causes and effects. The specific case of Minneapolis - Saint Paul (Twin Cities) metropolitan is examined using a panel of block-level land use and stop-level transit data. The results show that the development of land use, specifically, resident workers, can lead to the increase in bus demand, and thus further induce the increase in bus supply; the co-development of bus demand and supply is simultaneous on a yearly basis.

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1. Introduction

Transport and land use interactions have long been conceptualized by many researchers as a positive feedback system wherein urban form has substantial effects on the shape of transport systems while transport facilities affect the distribution of land use (Kelly, 1994).

Substantial empirical efforts have been made to investigate whether land use (e.g., density of development, overall urban structure) affects the transport network (Estupinan & Rodriguez, 2008; Mohammed et al., 2006; Zhao et al., 2014), and how the growth of the transport network changes land use patterns (Akgüngör et al., 2011; Beyzatlar & Kustepeli, 2011; Da Silveira et al., 2011; Kasraian et al., 2016; Wang et al., 2018). A few bi-directional studies have been conducted (Atack et al., 2010; Badoe & Miller, 2000; Kasraian et al., 2016; King, 2011; Levinson, 2007; Xie & Levinson, 2009b).

However, the two systems evolve dynamically (Giuliano, 2004) and so need to be considered simultaneously (Kasraian et al., 2016). Otherwise, errors might occur, for example, the effect of land use on travel demand could be exaggerated if the changes on traffic supply are ignored (Peng et al., 1997).

Levinson (2007) looked at the co-evolution of population and rail station densities for London and reported that they were mutually reinforced. Atack et al. (2010) investigated the relationship between rail access and changes in population density, as well as the rate of urbanization, in the Midwest United States, using a county-level panel data set from 1850-1860. It concluded that railways followed the existing population and contributed to the development of cities. King (2011) used Granger Causality models to explore the co-development of subway station density and

residential and commercial areas in New York at the parcel level from 1910 to 1950 and found that the growth of the subway network followed residential and commercial development, but was not a precursor to the residential development.

These studies examined such bi-directional transport and land use interaction from a long-run perspective, typically, at an interval of 10 years, which could support decision-making for long-term development plans, e.g., railway network expansion, or commercial center construction. Short-run investigations, which help to disclose relative immediate effects given the changes on either land use or transport, have not been conducted to date, while such analyses could provide insights for the design of transport operational plans such as changes to bus routes or timetables.

Previous bi-directional studies of transport-land use co-evolution have only considered changes in transport network supply or land development (Atack et al., 2010; King, 2011; Levinson, 2007), but have never examined the change in transport demand.

This paper addresses these gaps and constructs a comprehensive framework to explore short-run interactions considering elements representing land use, defined by job and resident worker density at each census block, as well as the bus network defined by bus demand (boardings and alightings per stop) and supply (scheduled number of buses per stop). We apply it to the Minneapolis - Saint Paul (Twin Cities) metropolitan area.

The remainder of the paper is structured as follows: Section 2 summarizes our theories and hypotheses. Section 3 describes the methodology. The collected data are elaborated in Section 4. The results are explained and discussed in Section 5. Section 6 concludes the paper.

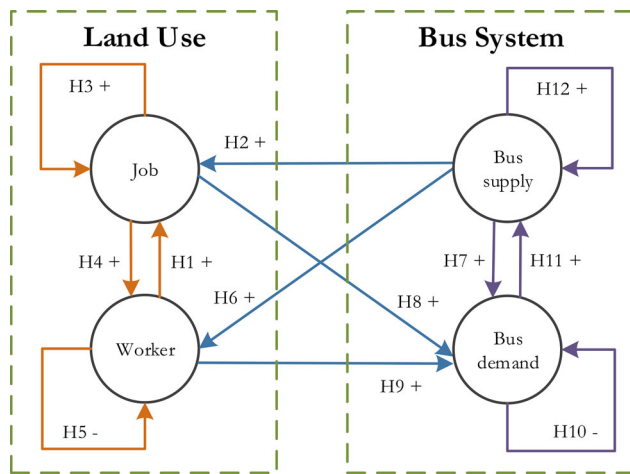


Figure 1. Hypotheses for interrelationships in land use and bus systems.

2. Theories and hypotheses

Urban economics highlights the basic tradeoff that workers want to have as much space as possible for the least cost while also desire to minimize travel time and costs and be near workplaces. The former can be thought of as the centrifugal force to drive workers apart and the latter as the centripetal force to drive them cluster in a region (Fujita et al., 1999). Economies of agglomeration predict that firms (jobs) want to be near each other, but they also want to be near their workforce. This is constrained because firms also want to have affordable space. So, one question would be whether economies of agglomeration are exhausted in any given metropolitan area at a particular stage of economic development and associated technologies, or remain salient in changing employment (job) locations, and whether diseconomies of agglomeration exceed the economies.

Bus supply and demand are the result of an ongoing process of balance, due to the induced travel effects (Peng et al., 1997). Increase in bus supply can improve service quality (Eboli & Mazzulla, 2011), leading to the rise of bus demand. Increased bus demand grows the profit (or political viability) of the bus system and so allows further the investment. When increasing bus supply, a traffic agency usually bundles stops and stations along a route extension, or increases frequency.

The spatial separation of land-use activities like home and work leads to interactions between different locations. Transport, such as a bus system, connect these separated activities. We hypothesize that land-use activities are thus attracted to locations with good bus services. When more activities are at a particular location, more demand for bus to and from the location are generated.

Therefore, we hypothesize the following, depicted in Figure 1, classified by corresponding dependent variables, as job density and worker density, and bus demand and bus supply, for further model specifications:

- Job density and worker density

H1: Job density is positively associated with the change of neighboring worker densities (Labor markets: employers want to be near workers).

H2: Job density is positively associated with the change of bus supply serving the block (Transit and land use feed-back loop, regional accessibility creates value).

H3: Job density has a positive relationship with the change of neighboring job densities (Economies of agglomeration, local accessibility creates value).

H4: Worker density is positively associated with the change of neighboring job densities (Labor markets: workers want to be near employment; local accessibility creates value).

H5: Worker density is negatively associated with the change of neighboring worker densities (Workers want to have more space and be away from other workers).

H6: Worker density is positively affected by the change of bus supply serving the block (Transit and land use feed-back loop, regional accessibility creates value).

- Bus demand and bus supply

H7: Bus demand (boardings/alightings) is positively associated with the change of bus supply (Induced demand, regional accessibility).

H8: Boardings/alightings are positively associated with the change of nearby jobs (Direct demand, local accessibility).

H9: Boardings/alightings are positively affected by the change of workers. (Direct demand, local accessibility).

H10: Boardings/alightings have negative relationships with the change of neighboring boardings/alightings. (Spatial dependency, substitution and competition effects).

H11: Bus supply is positively correlated with the change of bus demand (Induced supply).

H12: Bus supply is also positively correlated with the change of neighboring bus supplies. (Synergistic effect, network complementarity).

3. Methodology

3.1. Model specification

Granger Causality, proposed by Granger (1969), determines causality among related variables by checking whether one time series is statistically significant in forecasting another using F-tests. It interprets the strength of a relationship between the presence of one time series and the happening of another.

Granger Causality models are used here to investigate the cause-effect interactions between the developments of bus and land systems, following some literature with similar objectives (Binet, 2003; Cervero & Hansen, 2002; Fulton et al., 2000; Levinson, 2007; Xie & Levinson, 2009a). See specific details as follows.

3.1.1. Job density and worker density

Two models are specified to predict job density and worker density, respectively.

As the built environment changes slowly, we expect that the density at the present time can be largely explained by that in the previous time period, so lagged dependent variables are included to reduce auto-correlation in regression models (Levinson, 2007). The natural log transformations of

density variables consistently provide better fits than untransformed ones, so are selected.

Just as there is potential temporal auto-correlation in the panel data, the data also have a very well defined spatial structure, in this case, each block adjacent to some other blocks. Spatial variables of both job density and worker density are incorporated in each model, as we anticipate that they are mutually affected in space, by the state of each block in the nearby neighborhood, to capture the spatial effects (Tobler, 1970). The neighborhood of a block is specifically defined here as all other blocks, i.e. excluding itself, whose centroids are within the 400-meter buffer of its centroid.

The change of bus supply serving a block is included in the models to test the hypotheses of land use and transport cycle. We define that a bus stop serves a block if the stop is within the 400 m buffer of the block's centroid (other buffers were tested in the preliminary analysis, but a 400 m catchment provides the best fit). Stop catchment areas are typically based on walking access (Kuby et al., 2004). Some studies also assume that the maximal accessible walking distance is 400 m (Mohammed et al., 2006; Zhao et al., 2014).

To test the simultaneity of land use and bus systems, current change is used for all variables.

The general model can be represented as:

$$\ln(\rho_{i,t}) = \theta_0 + \theta_1 \ln(\rho_{i,t-k}) + \theta_2 \ln\left(\frac{\mathbf{W}_i * \mathbf{e}_t}{\mathbf{W}_i * \mathbf{e}_{t-k}}\right) + \theta_3 \ln\left(\frac{\mathbf{W}_i * \mathbf{r}_t}{\mathbf{W}_i * \mathbf{r}_{t-k}}\right) + \theta_4 \Delta s_{i,t,t-k} + \varepsilon \quad (1)$$

where:

$\rho_{i,t}$ is the land use density, specifically, job (e) or worker (w) density, at block i in year t ;

k is the lag length;

$\mathbf{W}_i$ is a spatial matrix identifying adjacency of block i ;

$\mathbf{e}_t$ and $\mathbf{r}_t$ are two vectors which, respectively, reflects job densities and resident worker densities of all blocks;

$\Delta s_{i,t,t-k}$ is the change of bus supply serving block i ;

ε is the random error.

3.1.2. Bus demand and bus supply

To explain the bus system changes, we build three models to explicitly explain its elements: boardings, alightings and bus supply at each stop.

Again, bus supply and demand are expected to be largely determined by the previous period as change occurs slowly, so the lagged variables are applied in the models accordingly. For the model fit, we found that the natural log transformation is better than no transformation as well. Hence, all of data are transformed here. The current change in bus supply is used to interpret bus demand, and vice versa, to examine the hypotheses of induced demand and induced supply. Additionally, to capture the spatial interdependence, we incorporate the neighboring change in every model, e.g., the change of neighboring bus supplies is specifically used, in which the neighborhood of a stop refers to those stops within the 400 m buffer of that stop, excluding itself.

Land use is expected to influence bus demand. So, similarly, the current changes of land use within the 400 m circle buffer centered at a stop (the number of jobs and workers) are used to explain its bus demand.

The models for bus demand and supply, respectively, are generalized as:

$$\begin{aligned} \ln(d_{j,t}) &= \beta_0 + \beta_1 \ln(d_{j,t-k}) + \beta_2 \ln\left(\frac{\mathbf{W}_j * \mathbf{d}_t}{\mathbf{W}_j * \mathbf{d}_{t-k}}\right) \\ &\quad + \beta_3 \Delta s_{j,t,t-k} + \beta_4 \Delta e_{j,t,t-k} + \beta_5 \Delta r_{j,t,t-k} + \varepsilon \\ \ln(s_{j,t}) &= \beta_6 + \beta_7 \ln(s_{j,t-k}) + \beta_8 \ln\left(\frac{\mathbf{W}_j * \mathbf{s}_t}{\mathbf{W}_j * \mathbf{s}_{t-k}}\right) \\ &\quad + \beta_9 \Delta d_{j,t,t-k} + \varepsilon \end{aligned} \quad (2) \quad (3)$$

where:

$d_{j,t}$ is the demand at stop j in year t , which can be specified to boardings (b) or alightings (a);

$s_{j,t}$ is bus supply at stop j in year t ;

$\mathbf{W}_j$ is a spatial weight matrix identifying adjacency of stop j ;

$\mathbf{d}_t$ and $\mathbf{s}_t$ are vectors which reflect bus demands and bus supplies of all stops, respectively;

$\Delta s_{j,t,t-k}$ is the change of bus supply at stop j from year $t-k$ to t ;

$\Delta e_{j,t,t-k}$ is the change of jobs within the catchment area of stop j ;

$\Delta w_{j,t,t-k}$ is the change of resident workers;

$\Delta d_{j,t,t-k}$ is the change of bus demand;

Other variables are the same as above.

3.2. Panel regression estimators

For cross-sectional time series (panel) data regression, a number of estimators have been proposed. Three widely used are feasible generalized least-squares (FGLS) estimator, panel-corrected standard error (PCSE) estimator, and non-parametric covariance matrix estimator because they adjust the standard errors of the coefficient estimates for time and spatial dependencies in the residuals, both of which are presented in many empirical applications (Hoechle, 2007; Jonsson, 2005). However, the FGLS estimator has a fatal disadvantage that it cannot be applied where the number of time periods is less than the number of cross-units. In addition, the method poorly performs in finite samples. The PCSE estimator, proposed by Beck and Katz (1995), corrected such problems, but requires assuming the form of spatial dependence correction, because the growth rate of the cross-section correlations is higher than that of observations (N^2 vs N), which is prone to bring errors in estimation. To further mitigate the problem, Driscoll and Kraay (1998) developed a nonparametric time-series covariance matrix estimator that applies a Newey-West-type correction to the sequence of cross-sectional averages of the moment conditions. It is consistent, independently of the cross-sectional dimension and thus appropriate for panel data with medium- and large-cross sectional dimension (Feld et al., 2020; Hoechle, 2007; Sarkodie & Strezov, 2019). For the

Table 1. Trend in the total number of jobs, workers, boardings and bus supply for Twin Cities.

Year	Jobs	Resident Workers	Boardings	Bus supply
2013	1,634,705	1,489,485	269,704	7,419
2014	1,662,813	1,517,995	248,011	7,572
2015	1,707,388	1,563,775	238,434	7,407
2016	1,734,378	1,583,645	225,645	7,568
2017	1,763,012	1,607,477	216,130	7,394

sake of generality, the nonparametric time-series covariance matrix estimator was selected. Specifically, it is executed in the Stata software program by the xtsc command.

4. Data

The Twin Cities metropolitan area is selected as a specific case and relevant data accordingly are collected to test the proposed hypotheses shown in Section 2.

The LEHD Original-Destination Employment Statistics (LODES) dataset, released by the US Census Bureau (2017), shows census block-level employment statistics from 2002 to 2017, in which the Workplace Area Characteristic (WAC) table provides the number of employees working in each census block, while the Residence Area Characteristic (RAC) table describes the number of workers living in each residential block. The data for the Twin Cities is extracted and used to track the changes of land use distributions.

Stop-level micro bus data by route and direction is collected from Metropolitan Council and Metro Transit (2012), in which boardings, alightings and the number of buses are averaged on a daily basis from the year 2013 to 2019. Light rail data are not included in this study due to the lack of data in the year 2013. We further accumulate the data by stop to track the spatio-temporal variations of bus systems. Note that stop-level boarding and alighting records are only available for bus services provided by Metro Transit, which is the core of bus system in the Twin Cities and carries about 85% of the region's transit passengers (Metro Transit & Metropolitan Council, 2018), so are selected for this analysis.

To investigate the interaction between land use and bus systems, the overlapped period, from 2013 to 2017, is selected. Table 1 summarizes the data and depicts the changes over time. The uptrend can be clearly observed for the number of jobs and workers due to the increase of population and the growth of economy, while bus demand is falling, especially from 2013 to 2014, which is attributed to the increasing dependency on automobile led by the growth of economy and the mid 2014-opening of the Green Line light rail that replaces selected bus services. Bus supply fluctuates, as the transit agency aims to optimize the network to align with and shape bus demand.

It should be noted that all else equal, the number of jobs or workers at each block and bus demand (boardings or alightings) and supply at each stop follow the system trend. When the economy blooms, the total number of jobs increases, the regional number increases, and the number of jobs everywhere increase on average. When it withers, they all decrease. Similarly, the number of workers at each block

Table 2. Predicting job and worker density.

Variables	ln($e_{i,t}$)				ln($r_{i,t}$)			
	Hypo.	Est.	Drisc/Kraay S.E.	Sig.	Hypo.	Est.	Drisc/Kraay S.E.	Sig.
ln($e_{i,t-1}$)		0.963	0.004	***				
ln($w_{i,t-1}$)						0.888	0.022	***
$\Delta s_{i,t,t-1}$	H2 +	-0.046	0.205		H6 +	0.064	0.278	
ln($\frac{W \cdot e_{i,t}}{W \cdot e_{i,t-1}}$)	H3 +	-0.033	0.018		H4 +	0.003	0.002	
ln($\frac{W \cdot r_{i,t}}{W \cdot r_{i,t-1}}$)	H1 +	0.025	0.010		H5 -	0.194	0.066	
Const.		-0.286	0.034	***		-0.818	0.153	*
Number of obs.		25654				111935		
Number of groups		6741				28893		
R ²		0.923				0.806		
Prob > F		0.024				0.040		

*** $p < 0.005$, ** $p < 0.01$, * $p < 0.05$.

and boardings (alightings, bus supply) at each stop also have positive relationships with the total number. To control for the system trend, we normalize these data. Specifically,

1. the ratio of jobs (workers) at a block to regional total jobs (workers), i.e. the proportion of jobs (workers) is used rather than the number of jobs (workers) directly.
2. the ratio of boardings (alightings, bus supply) at a stop to the total of boardings (alightings, bus supply), i.e. the proportion of boardings (alightings, bus supply) rather than boardings (alightings, bus supply) is used.

Hence, from here on out, when we say *number*, we mean the ratio.

The TIGER/Line shapefile, defined by US Census Bureau and Metropolitan Council (2012) with selected geographic and cartographic information, can be joined with the LODES dataset to connect land use information with geometry features using unique census block identifiers. The transit stops shapefile, released by Metropolitan Council and Metro Transit (2018), is then used to locate the stops. Given that, we establish the spatial relationships between blocks and stops.

5. Results

In all, five models are presented: job density and worker density results are given in Table 2, and boardings, alightings, and bus supply results are shown in Table 3. The elasticities of all significant variables are displayed in Table 4, which are evaluated at the corresponding mean value points.

5.1. Job density and worker density

Not surprisingly, density at the current time is positively explained by one-period lagged density. None of the other variables, however, are statistically significant, so hypotheses H1-H6 are not corroborated.

Overall, these results suggest that the changes of both neighboring land use intensities and bus supplies do not immediately drive the changes in land use intensities in the

Table 3. Predicting boardings, alightings and bus supply.

Variables	$\ln(b_{j,t})$				$\ln(a_{j,t})$				$\ln(s_{j,t})$			
	Hypo.	Est.	Drisc/Kraay S.E.	Sig.	Hypo.	Est.	Drisc/Kraay S.E.	Sig.	Hypo.	Est.	Drisc/Kraay S.E.	Sig.
$\ln(b_{j,t-1})$		0.934	0.004	***								
$\ln(a_{j,t-1})$					0.942	0.003		***				
$\ln(s_{j,t-1})$										0.989	0.003	***
$\Delta b_{j,t-1}$									H11 +	91.925	6.915	***
$\Delta a_{j,t-1}$									H11 +	146.039	23.619	**
$\Delta s_{j,t-1}$	H7 +	61.838	8.794	**	H7 +	64.609	9.156	**				
$\Delta e_{j,t-1}$	H8 +	-17.600	7.410		H8 +	1.822	14.705					
$\Delta f_{j,t-1}$	H9 +	261.850	45.184	*	H9 +	174.816	46.122	*				
$\ln\left(\frac{W \cdot b_{j,t}}{W \cdot b_{j,t-1}}\right)$	H10 -	0.060	0.064									
$\ln\left(\frac{W \cdot a_{j,t}}{W \cdot a_{j,t-1}}\right)$					H10 -	0.102	0.053					
$\ln\left(\frac{W \cdot s_{j,t}}{W \cdot s_{j,t-1}}\right)$									H12 +	0.601	0.049	***
Const.		-0.657	0.048	***		-0.582	0.041	***		-0.061	0.015	*
Number of obs.			26243				27041				44132	
Number of groups			7304				7457				11037	
R ²			0.893				0.905				0.980	
Prob > F			0.001				0.016				0.000	

*** p < 0.005, ** p < 0.01, * p < 0.05.

Table 4. Elasticities for land use and bus systems.

Indep. Var.	Dep. Var.	Hypo.	Mean	Elasticity
$\Delta s_{j,t-1}$	$b_{j,t}$	H7 +	2.71E-05	1.68E-03
$\Delta r_{j,t-1}$	$b_{j,t}$	H9 +	5.37E-06	1.41E-03
$\Delta s_{j,t-1}$	$a_{j,t}$	H7 +	2.64E-05	1.71E-03
$\Delta r_{j,t-1}$	$a_{j,t}$	H9 +	5.21E-06	9.11E-04
$\Delta b_{j,t-1}$	$s_{j,t}$	H11 +	5.67E-07	5.21E-05
$\Delta a_{j,t-1}$	$s_{j,t}$	H11 +	5.35E-07	7.81E-05
$\frac{W \cdot s_{j,t}}{W \cdot s_{j,t-1}}$	$s_{j,t}$	H12 +		6.01E-01

Twin Cities in this period. It might be explained by the observation of spatial development history that people actually follow jobs (Storper, 2011) rather than react simultaneously. With regards to bus supply, many alternative modes, especially automobile, allow travelers to meet their need for reaching activities, so the importance of bus supply in location choice is diminished. Hence, future work needs to dig further the cause of the change in land use intensities, such as the lagged effect of land use intensities, other travel modes, zoning policies (Verburg et al., 2004) and topographic condition (Zhou & Kockelman, 2008).

5.2. Bus demand and bus supply

Turning to the question of the bus system, the results of models predicting boardings, alightings, and bus supply are summarized in Table 3. Overall, the bus system is strongly determined by its previous state, as the built environment changes slowly.

For each hypothesis individually:

H7: Boardings or alightings at a stop are positively associated with the change of bus supply, which corroborates the hypothesis of induced demand. A one percent increase in the change of bus supply increases the boardings by 0.168% (evaluated at mean value point of 0.000027) and the alightings by 0.171% (evaluated at mean value point of 0.000026).

H8: Jobs in a stop catchment area do not significantly affect the boardings or alightings of that stop, which demonstrates that the development of jobs is not a short-term factor driving bus demand.

H9: Resident workers in a stop catchment area positively and significantly affect the demand of that stop, which corroborates direct demand. The elasticities of the change in resident workers for boardings and alightings are 0.0014 and 0.0009 (at mean value point: 0.000005), respectively.

H10: The neighboring effects of boardings on boardings and alightings on alightings, are both insignificant. These reject the spatial auto-correlations of bus demands.

H11: The increase in bus demand leads to the increase of bus supply, which provides the evidence of induced supply. A 1% increase in the changes of boardings and alighting, respectively, lift bus supply by 0.005% (at mean value point: 0.00000057) and by 0.008% (at mean value point: 0.00000054), respectively.

H12: Bus supply at a stop is positively affected by the change of neighboring bus supplies, with an elasticity of 0.6. This supports the hypothesis of synergistic effect of bus services in space, i.e. the agency is clustering services to support the complementary bus system and allowing for transfers, rather than viewing stops as substitutes.

6. Conclusions

This research proposes a general framework to describe the potential external and internal interactions between land use and transit systems. This tests the general transport-land use interaction mode, which has had mixed empirical results in previous research. These interactions are tested for the case of the Twin Cities through a series of panel regression models using block-level land use data and stop-level transit data from 2013 to 2017 on a yearly basis.

The results do not show that significant immediate interactions between land use intensities (job and worker density). In addition, bus supply does not Granger-cause the

changes in land use intensities, suggesting that the simultaneous change of land use intensity and bus system is uncommon in the Twin Cities region.

With regard to bus system, there is a positive simultaneous feedback between bus demand and bus supply on a yearly basis that bus demand immediately leads to the expansion of bus supply and the development of bus supply also causes the immediate growth of bus demand. The increase in the number of resident workers in an area Granger-causes the growth of bus demand, but the increase of the number of jobs does not.

Overall, we can conclude that land development immediately leads to the changes in the bus system in the short-run. It directly affects bus demand and then indirectly affects bus supply by changing bus demand. But, in the case of Twin Cities, there is no significant effect of bus supply on land use over this period. In the short-run, unlike large physical infrastructure projects, there is induced supply for transit services (Chakraborty & Mishra, 2013; Dill et al., 2013; Sung et al., 2014), which are easy to deploy, but no evidence for induced development, corroborating the study of Benson and Cao (2020) of Rapid Bus services in St. Paul, Minnesota.

One generalizable implication arises from this research. Small investments in bus services do not necessarily change land use patterns or induce land development in the short run in places with heavy reliance on automobiles, as is true in the Minneapolis - St. Paul region.

We test all hypotheses using a 5-year-long dataset, which cannot track changes to land use or bus networks that take place over a longer time-horizon. Future research can address this as more data become available.

Note that, the statistical findings from these results are for the Twin Cities, while other cities may have different results considering the specific evolution of land use distributions and network structure forms. The framework identifying the potential interaction as well as the modeling approach, however, can be employed to detect other policy responses. Future research on other cities will establish the robustness of these results and their potential for transferability. Further analyses should also test the interaction considering other travel modes like railway, as well as other characteristics of public transport service, which may yield additional insights.

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Author statement

Manman Li: Methodology, Formal analysis, Writing- Original draft preparation; Mengying Cui: Methodology, Validation, Writing -

Reviewing and Editing; David Levinson: Conceptualization, Methodology, Writing - Reviewing and Editing, Supervision.

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Data availability statement

The data that support the findings of this study are openly available in <https://lehd.ces.census.gov/data/#lodes.html>; <https://gisdata.mn.gov/dataset/us-mn-state-metc-trans-stop-boardings-alightings>.

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